Zero-Error Identification and Rate-Query Tradeoffs in Classification Systems

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Abstract

We study zero-error class identification under constrained observations with three resources: tag rate L (bits per entity), identification cost W (attribute queries), and distortion D (misidentification probability). If the attribute-profile map π is not injective on classes, then attribute-only observation cannot identify class identity with zero error. Let $A_\pi := \max_u |\{c : \pi(c) = u\}|$ be collision multiplicity. Any $D = 0$ scheme must satisfy $L \geq \log_2 A_\pi$, and this bound is tight. In maximal-barrier domains ($A_\pi = k$), the nominal point $(L, W, D) = (\lceil \log_2 k \rceil, O(1), 0)$ is the unique Pareto-optimal zero-error solution under the stated observation model: no other feasible combination achieves $D = 0$. Without tags ($L = 0$), zero-error identification requires $W = \Omega(d)$ queries, where d is the distinguishing dimension. Minimal sufficient query sets form the bases of a matroid, making d well-defined as an invariant of the observation structure rather than of a particular query strategy. We also prove exact finite-block confusability laws: at blocklength t, zero-error feasibility is equivalent to a tag alphabet threshold A_π^t, yielding exact linear log-bit scaling. Conceptually, the framework extends classical rate-distortion accounting by treating query cost as an additional interactive resource. We then study robustness in open worlds, implementation-level maintenance consequences, and cross-domain instantiations across databases, biology, software runtimes, and model registries. The claims are accompanied by supplementary machine verification in Lean 4 (8092 lines, 352 theorem/lemma statements, 0 `sorry`).

Keywords: rate-distortion theory, zero-error identification, confusability graphs, query complexity, matroid structure, classification systems

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I. Introduction

A. The Identification Problem

Consider an encoder-decoder pair communicating about entities from a large universe $\mathcal{V}$. The decoder must *identify* each entity, determining which of k classes it belongs to, using only:

- a *tag* of L bits stored with the entity, and/or
- *queries* to a binary oracle: "does entity v satisfy attribute I?"

This is not reconstruction (the decoder need not recover v), but *identification* in the sense that the decoder must answer "which class?" with zero or bounded error. Our work extends this setting to track the tradeoff between tag storage, query complexity, and identification accuracy under constrained observation.

The observational bottleneck is the attribute-profile map $\pi : \mathcal{V} \to \{0, 1\}^n$. Once distinct classes collide under π, attribute-only observation cannot distinguish them with zero error. The paper turns that obstruction into a compression law. Let

$$A_\pi := \max_u |\{c : \pi(c) = u\}|$$

be the largest collision block. Then every zero-error scheme must satisfy

$$L \geq \log_2 A_\pi,$$

and this lower bound is tight. The same law extends exactly to finite blocks: the minimum block tag budget is $L_t^\star = t \log_2 A_\pi$.

We then develop a second theorem arc for the query resource. Without tags ($L = 0$), zero-error identification depends on the structure of distinguishing query families. Minimal sufficient query sets form the bases of a matroid, so the distinguishing dimension d is well-defined as an invariant of the observation structure rather than of a particular query strategy. Every tag-free zero-error scheme therefore has worst-case query cost at least d.

These laws are domain-independent within the stated model: the same lower bounds appear across databases, biological taxonomy, knowledge graphs, model registries, and software runtimes. Once classes collide under observable attributes, constant-cost zero-error identification requires auxiliary naming information. In maximal-barrier domains this yields a unique $D = 0$ Pareto solution, so the issue is resource feasibility rather than domain-specific style conventions.

B. Main Results

We organize the paper around one primary theorem arc and one secondary consequences arc.

1) **Zero-error ambiguity converse and exact scaling.** Any $D = 0$ scheme must satisfy $L \geq \log_2 A_\pi$ (Theorem III.13), and the finite-block confusability law gives exact scaling $L_t^\star = t \log_2 A_\pi$ (Corollary V.7). In maximal-barrier domains ($A_\pi = k$), the nominal point $(\lceil \log_2 k \rceil, O(1), 0)$ is the unique Pareto-optimal zero-error solution (Theorem V.1).
2) **Query lower bound and matroid structure.** Minimal distinguishing query sets form the bases of a matroid (Theorem IV.6). Their common cardinality d is the distinguishing dimension (Definition IV.7), and any attribute-only zero-error scheme has worst-case query cost at least d (Theorem IV.10). The matroid theorem is what makes d strategy-independent rather than a proof artifact.
3) **Foundational observational impossibility.** The information barrier theorem (Theorem II.9) shows that attribute-only observers cannot compute properties that vary within observational equivalence classes. This is the base invariance statement from which the converse, the Pareto characterization, and the query lower bounds follow.
4) **Open-world robustness and decidability boundary.** Barrier-freedom is not extension-stable in open worlds (Theorem VI.3), and for arbitrary generators it is not computably certifiable in advance (Theorem VI.5). A persistent zero-error guarantee therefore cannot rely on present-day collision-freedom alone.
5) **Implementation-level consequences.** When identity is carried explicitly, error localization and constraint centralization collapse to $O(1)$ sites; when systems rely on distributed attribute probes, localization and information scattering scale with the number of probe sites (Section VII).

6) **Machine-checked proofs.** The claims are accompanied by a supplementary Lean 4 artifact (8092 lines, 352 theorem/lemma statements, 0 `sorry`). The mechanization supports the mathematics rather than acting as a separate theorem-family contribution.

(L: ROB1, ROB2, UND1, UND2)

Without the equicardinality result behind the matroid structure, the lower bound would depend on which distinguishing set or query strategy one chose. The matroid theorem is what makes d strategy-independent.

C. Related Work and Positioning

This paper does not develop asymptotic coding theorems or claim new zero-error capacity results. Its contribution is to make explicit the bit/query/error tradeoffs governing zero-error classification under constrained observations.

Identification via channels. Our work is adjacent to the identification paradigm introduced by Ahlswede and Dueck [1], [2]. In their framework, a decoder need not reconstruct a message but only answer "is the message m?" for a given hypothesis. Our setting is different in three ways: (1) we consider one-shot zero-error class identification rather than asymptotic vanishing-error identification, (2) queries are adaptive rather than block codes, and (3) we allow auxiliary tagging (rate L) to reduce query cost. The connection is motivational and terminological rather than a direct reuse of the original coding theorem.

Rate-distortion theory. The (L, W, D) framework is analogous to rate-distortion theory [3], [4]: the "distortion" D is semantic (class misidentification), and there is a second resource W (query cost) alongside rate L. Classical rate-distortion asks for the minimum rate needed to achieve distortion D; here the main question is which combinations of tag bits and queries permit $D = 0$. Theorem V.1 gives the resulting converse in terms of collision multiplicity and identifies the unique nominal point in the maximal-barrier regime.

Rate-distortion-perception tradeoffs. Blau and Michaeli [5] extended rate-distortion theory by adding a perception constraint, creating a three-way tradeoff. Our query cost W plays an analogous role: it measures the interactive cost of achieving low distortion rather than a distributional constraint. We cite this as a structural analogy, not as a claim that the present model inherits the full geometry of that setting.

Zero-error information theory. Körner [6] and Witsenhausen [7] studied zero-error source coding where confusable symbols must be distinguished. Here the direct theorem is simpler: the ambiguity converse yields an explicit minimum zero-error tag budget, and the paper also proves exact finite-block confusability laws with linear log-bit scaling. These strengthen the operational converse while remaining distinct from full distribution-optimized graph-entropy or capacity theorems.

Query complexity and communication complexity. The $\Omega(d)$ lower bound for attribute-only identification relates to decision tree complexity [8] and interactive communication [9]. The key distinction is that our queries are constrained to a fixed observable family $\mathcal{I}$, not arbitrary predicates.

Compression in classification systems. The same ambiguity bound appears across databases, knowledge graphs, taxonomy, and software-runtime typing systems: for zero-error identification, ambiguity induces a minimum metadata requirement $L \geq \log_2 A_\pi$ (Theorem III.13), with maximal-barrier specialization $L \geq \log_2 k$.

Runtime or schema instantiation (secondary). In nominal-vs-structural interface settings [10], [11], the model yields a concrete cost statement: under attribute collisions, purely profile-based identification has worst-case $\Omega(d)$ witness cost, while explicit tags achieve $O(1)$ identification using $O(\log A_\pi)$ bits. This is one instantiation of the general bounds.

D. Paper Organization

Section II formalizes the observation model and the foundational information barrier. Section III develops one-shot zero-error feasibility, nominal tagging, and the ambiguity-based converse. Section IV proves the matroid structure of distinguishing query families. Section V gives the (L, W, D) frontier and exact finite-block scaling. Section VI studies open-world robustness and the decidability boundary. Section VII collects practical consequences and cross-domain illustrations. Section **??** sketches extensions and open directions. Section VIII concludes.

II. Model and Observational Impossibility

A. Semantic Compression: The Problem

The fundamental problem of *semantic compression* is: given a value v from a large space $\mathcal{V}$, how can we represent v compactly while preserving the ability to answer semantic queries about v? This differs from classical source coding in that the goal is not reconstruction but *identification*: determining which equivalence class v belongs to.

Classical rate-distortion theory [3] studies the tradeoff between representation size and reconstruction fidelity. We extend this to a discrete classification setting with three dimensions: *tag length* L (bits of storage), *witness cost* W (queries or bits of communication required to determine class membership), and *distortion* D (misclassification probability).

This work treats contemporary classification problems with explicit resource accounting. The point is not to claim a new Shannon-style asymptotic coding theorem, but to make explicit the bit/query/error tradeoffs that already constrain zero-error identification in practice.

B. Universe of Discourse

Definition II.1 (Entity space and attribute family)**.** Let $\mathcal{V}$ be a set of entities (program objects, database records, biological specimens, library items). Let $\mathcal{I}$ be a finite set of binary *attributes*. Each $I \in \mathcal{I}$ induces a bipartition of $\mathcal{V}$ via q_I, and the full family induces the observational equivalence partition.

(L: ACS5, COMPL1)

Remark II.2 (Terminology). We use "attribute" for the abstract concept. Depending on domain, attributes may be interfaces/signatures, database columns, phenotypic characters, or catalog facets. The mathematics is identical.

Definition II.3 (Attribute observation family)**.** For each $I \in \mathcal{I}$, define the attribute-membership observation $q_I : \mathcal{V} \to \{0,1\}$:

$$q_I(v) = \begin{cases} 1 & \text{if } v \text{ satisfies attribute } I \\ 0 & \text{otherwise} \end{cases}$$

Let $\Phi_{\mathcal{I}} = \{q_I : I \in \mathcal{I}\}$ denote the attribute observation family.

(L: ACS5, COMPL1)

Remark II.4 (Notation for size parameters). We write $n := |\mathcal{I}|$ for the ambient number of available attributes. We write d for the distinguishing dimension (the common size of all minimal distinguishing query sets; Definition IV.7), so $d \leq n$ and there exist worst-case families with $d = n$. We write m for the number of *query sites* (call sites) that perform attribute checks in a program or protocol (used only in the complexity-of-maintenance discussion). When discussing a particular identification or verification task, we may write s for the number of attributes actually queried or traversed by the procedure, with $s \leq n$.

Definition II.5 (Attribute profile)**.** The attribute profile function $\pi : \mathcal{V} \to \{0,1\}^{|\mathcal{I}|}$ maps each value to its complete attribute signature:

$$\pi(v) = (q_I(v))_{I \in \mathcal{I}}.$$

(L: ACS8, COMPL1)

Definition II.6 (Attribute indistinguishability)**.** Values $v, w \in \mathcal{V}$ are *attribute-indistinguishable*, written $v \sim w$, iff $\pi(v) = \pi(w)$.

(L: ACS8, ACS9)

The relation $\sim$ is an equivalence relation. We write $[v]_\sim$ for the equivalence class of v.

Definition II.7 (Classification scheme)**.** A *classification scheme* is any procedure (deterministic or randomized), with arbitrary time and memory, whose only access to a value $v \in \mathcal{V}$ is via:

1) the observation family $\Phi = \{q_I : I \in \mathcal{I}\}$, where $q_I(v) = 1$ iff v satisfies attribute I, and optionally

2) a nominal-tag primitive $\tau : \mathcal{V} \to \mathcal{T}$ returning an opaque class identifier.

All theorems in this paper quantify over all such schemes.

(L: ACS5, COMPL1, FORC2)

This definition is intentionally broad: schemes may be adaptive, randomized, or computationally unbounded. The constraint is *observational*, not computational.

Definition II.8 (Attribute-only observer)**.** An *attribute-only observer* is any procedure whose interaction with a value $v \in \mathcal{V}$ is limited to queries in $\Phi_{\mathcal{I}}$. Formally, the observer interacts with v only via primitive attribute queries $q_I \in \Phi_{\mathcal{I}}$; hence any transcript (and output) factors through $\pi(v)$.

(L: ACS8, COMPL1)

C. The Central Question

The central question is: **what semantic properties can an attribute-only observer compute?**

A semantic property is a function $P : \mathcal{V} \to \{0,1\}$ (or more generally, $P : \mathcal{V} \to Y$ for some codomain Y). We say P is *attribute-computable* if there exists a function $f : \{0,1\}^{|\mathcal{I}|} \to Y$ such that $P(v) = f(\pi(v))$ for all v.

D. The Information Barrier

Theorem II.9 (Information barrier)**.** *Let $P : \mathcal{V} \to Y$ be any function. If P is attribute-computable, then P is constant on $\sim$-equivalence classes:*

$$v \sim w \implies P(v) = P(w).$$

Equivalently: no attribute-only observer can compute any property that varies within an equivalence class.

(L: ACS8, ACS9, COH1)

Proof. Suppose P is attribute-computable via f, i.e., $P(v) = f(\pi(v))$ for all v. Let $v \sim w$, so $\pi(v) = \pi(w)$. Then:

$$P(v) = f(\pi(v)) = f(\pi(w)) = P(w).$$

■

Remark II.10 (Information-theoretic nature). The barrier is *informational*, not computational. Given unlimited time, memory, and computational power, an attribute-only observer still cannot distinguish v from w when $\pi(v) = \pi(w)$. The constraint is on the evidence itself.

Remark II.11 (Role in the paper). Theorem II.9 is the base invariance statement. The technical contribution is the downstream structure built on top of it: the ambiguity-based converse (Theorem III.13), the Pareto characterization (Theorem V.1), and the matroid or equicardinality results (Section IV).

Corollary II.12 (Class identity is not attribute-computable)**.** *Let $C : \mathcal{V} \to \{1, \ldots, k\}$ be the class assignment function. If there exist values v, w with $\pi(v) = \pi(w)$ but $C(v) \neq C(w)$, then class identity is not attribute-computable.*

(L: ACS9)

Proof. Direct application of Theorem II.9 to $P = C$. ■

E. Model Capture and Fixed-Axis Domains

Proposition II.13 (Model capture)**.** *Any real-world classification protocol whose evidence consists solely of attribute-membership queries is representable as a scheme in the above model. Conversely, any additional capability corresponds to adding new observations to Φ.*

(L: ACS5, COMPL1, FORC2)

This proposition forces any objection into a precise form: to claim the theorem does not apply, one must name the additional observation capability not in Φ. "Different universe" is not a coherent objection; it must reduce to "I have access to oracle $X \notin \Phi$."

The core results require only $(\mathcal{V}, C, \pi)$ and the observation family Φ. The two-axis decomposition below is one representative instantiation and is not an additional axiom for the general theorems.

In that instantiation, each value is characterized by:

- **Provenance axis** (**B**): source or lineage information that can distinguish entities with identical observable profiles[1]
- **Profile axis** (**S**): the observable attribute profile induced by Φ

Definition II.14 (Two-axis representation)**.** A value $v \in \mathcal{V}$ has representation $(B(v), S(v))$ where:

$$B(v) = \text{provenance}(v) \tag{1}$$

$$S(v) = \pi(v) = (q_I(v))_{I \in \mathcal{I}}. \tag{2}$$

The provenance axis captures *identity-bearing origin information*; the profile axis captures *observable behavior or attributes*.

In concrete deployments, B may be carried by lineage metadata, registry identifiers, or equivalent provenance carriers. Any implementation-specific normalization or lookup machinery is auxiliary.

(L: ACS5, DOM1, FORC2)

Theorem II.15 (Fixed-axis completeness)**.** *Let a fixed-axis domain be specified by an axis map $\alpha : \mathcal{V} \to \mathcal{A}$ and an observation family Φ such that each primitive query $q \in \Phi$ factors through α. Then every in-scope semantic property (i.e., any property computable by an admissible Φ-only strategy) factors through α: there exists $\tilde{P}$ with*

$$P(v) = \tilde{P}(\alpha(v)) \quad \textit{for all } v \in \mathcal{V}.$$

(L: ACS5, COMPL1, DOM1)

[1] In the Lean witness model, this axis is represented as `Bases`.

In this two-axis instantiation, $\alpha(v) = (B(v), S(v))$, so in-scope semantic properties are functions of (B, S).

Proof. An admissible Φ-only strategy observes v solely through responses to primitive queries $q_I \in \Phi$. By hypothesis each such response is a function of $\alpha(v)$. Therefore every query transcript, and hence any strategy's output, depends only on $\alpha(v)$, so the computed property factors through α. ■

Corollary II.16 (Fixed-axis incompleteness)**.** *Any fixed-axis classification system is complete only for properties measurable on the fixed axis map α, and incomplete for any property that varies within an α-fiber. Equivalently, if $\alpha(v) = \alpha(w)$ but $P(v) \neq P(w)$, then no admissible Φ-only strategy can compute P with zero error.*

(L: L2)

Proposition II.17 (Observational quotient)**.** *For any admissible strategy using only Φ, the entire interaction transcript (and hence the output) depends only on $\alpha(v)$. Equivalently, any in-scope semantic property P factors through α: there exists $\tilde{P}$ with $P(v) = \tilde{P}(\alpha(v))$ for all v.*

(L: ACS8, ACS9, COH1)

Corollary II.18 (Why "ad hoc" = adding an axis or tag)**.** *If two values v, w satisfy $\alpha(v) = \alpha(w)$, then no admissible Φ-only strategy can distinguish them with zero error. Any mechanism that does distinguish such pairs must introduce additional information not present in α (equivalently, refine the axis map by adding a new axis or tag).*

(L: ACS8, COMPL1, FORC2)

F. Attribute Equivalence and Observational Limits

Proposition II.19 (Equivalence class structure)**.** *The relation $\sim$ partitions $\mathcal{V}$ into equivalence classes. Let $\mathcal{V}/\sim$ denote the quotient space. An attribute-only observer effectively operates on $\mathcal{V}/\sim$, not $\mathcal{V}$.*

(L: ACS8, COMPL1, FORC2)

Corollary II.20 (Information loss quantification)**.** *The information lost by attribute-only observation is:*

$$H(\mathcal{V}) - H(\mathcal{V}/\sim) = H(\mathcal{V} \mid \pi)$$

where H denotes entropy. This quantity is positive whenever multiple classes share the same attribute profile.

(L: ACS8)

III. Zero-Error Identification and Tag-Rate Lower Bounds

A. One-Shot Zero-Error Feasibility

We now formalize the identification problem in channel-theoretic terms. Let $C : \mathcal{V} \to \{1, \ldots, k\}$ denote the class assignment function, and let $\pi : \mathcal{V} \to \{0, 1\}^n$ denote the attribute profile.

Definition III.1 (Observation channel)**.** The *observation channel* induced by observation family Φ is the mapping $C \to \pi(V)$ for a random entity V drawn from distribution P_V over $\mathcal{V}$. The channel output is the attribute profile; the channel input is implicitly the class $C(V)$.

Theorem III.2 (One-Shot Zero-Error Identification Feasibility)**.** *Let $\mathcal{C} = \{1, \ldots, k\}$ be the class space. Zero-error identification of all k classes is achievable if and only if every class has a distinct attribute profile. When π is not class-injective, no zero-error identification scheme exists without auxiliary tag information.*

(L: ACS8)

Proof. Achievability: If π is injective on classes, then observing $\pi(v)$ determines $C(v)$ uniquely. The decoder simply inverts the class-to-profile mapping.

Converse: Suppose two distinct classes $c_1 \neq c_2$ share a profile: there exist $v_1 \in c_1, v_2 \in c_2$ with $\pi(v_1) = \pi(v_2)$. Then any decoder $g(\pi(v))$ outputs the same class label on both v_1 and v_2, so it cannot be correct for both. Hence zero-error identification of all classes is impossible. ■

Remark III.3 (Information-theoretic corollary). Under any distribution with positive mass on both colliding classes, $I(C; \pi(V)) < H(C)$. This is an average-case consequence of the deterministic barrier above.

(L: ACS8)

Remark III.4 (Relation to Ahlswede-Dueck). In the identification paradigm of [1], the decoder asks "is the message m?" rather than "what is the message?" This yields double-exponential codebook sizes. Our setting is different: we require zero-error identification of the *class*, not hypothesis testing. The one-shot zero-error identification feasibility threshold is binary rather than asymptotic.

(L: ACS8)

Remark III.5 (Terminology). Theorem III.2 is a one-shot feasibility statement, not a Shannon asymptotic coding theorem. The terminology is retained only as compact shorthand for the zero-error or non-zero-error dichotomy under the stated observation model.

(L: ACS8)

B. The Positive Result: Nominal Tagging

We now show that augmenting attribute observations with a single primitive, nominal-tag access, achieves constant witness cost.

Definition III.6 (Nominal-tag access)**.** A *nominal tag* is a value $\tau(v) \in \mathcal{T}$ associated with each $v \in \mathcal{V}$, representing the class identity of v. The nominal-tag access operation returns $\tau(v)$ in $O(1)$ time.

(L: ACS7)

Definition III.7 (Primitive query set)**.** The extended primitive query set is $\Phi_{\mathcal{I}}^{+} = \Phi_{\mathcal{I}} \cup \{\tau\}$, where τ denotes nominal-tag access.

(L: FORC2)

Theorem III.8 (Tag-Restored Zero Error (Sufficiency))**.** *A class-injective nominal tag of length $L \geq \lceil \log_2 k \rceil$ bits suffices for zero-error identification, regardless of whether π is class-injective.*

(L: ACS7)

Proof. A nominal tag $\tau : \mathcal{V} \to \{1, \ldots, k\}$ assigns a unique identifier to each class. Reading $\tau(v)$ determines $C(v)$ in $O(1)$ queries, independent of the attribute channel. ∎

Definition III.9 (Witness cost)**.** Let $W(P)$ denote the minimum number of primitive queries from $\Phi^+_{\mathcal{I}}$ required to compute property P. We distinguish two tasks:

- W_{id}: Cost to identify the class of a single entity.
- W_{eq}: Cost to determine if two entities have the same class.

Unless specified, W refers to worst-case identification cost.

(L: ACS1, ACS7)

Theorem III.10 (Constant witness for class identity)**.** *Under nominal-tag access, class identity checking has constant witness cost:*

$$W(\textit{class-identity}) = O(1).$$

Specifically, the witness procedure is to compare the relevant tag values.

(L: ACS7, FORC2, DOM1)

Proof. The procedure makes a constant number of primitive queries (one tag access per value) and one comparison. This is $O(1)$ regardless of the number of available attributes $|\mathcal{I}|$. ∎

C. Converse: Tag Rate Lower Bound

Definition III.11 (Collision multiplicity)**.** Let $\mathcal{C} = \{1, \ldots, k\}$ and let $\pi_{\mathcal{C}} : \mathcal{C} \to \{0,1\}^n$ be the class-level profile map. Define

$$A_\pi := \max_{u \in \mathrm{Im}(\pi_{\mathcal{C}})} |\{c \in \mathcal{C} : \pi_{\mathcal{C}}(c) = u\}| \,.$$

Thus A_π is the size of the largest class-collision block under observable profiles.

(L: LWDC1, LWDC3)

Definition III.12 (Maximal-Barrier Regime)**.** The domain is *maximal-barrier* if $A_\pi = k$, i.e., all classes collide under the observation map.

(L: LWDC3)

Theorem III.13 (Converse)**.** *For any classification domain, any scheme achieving $D = 0$ requires*

$$2^L \geq A_\pi \quad \textit{equivalently} \quad L \geq \log_2 A_\pi,$$

where A_π is the collision multiplicity.

(L: LWDC1)

Proof. Fix a collision block $G \subseteq \mathcal{C}$ with $|G| = A_\pi$ and identical observable profile. For classes in G, query transcripts are identical, so zero-error decoding must separate those classes using tag outcomes. With L tag bits there are at most 2^L outcomes, hence $2^L \geq |G| = A_\pi$. ■

Corollary III.14 (Maximal-Barrier Converse)**.** *If the domain is maximal-barrier ($A_\pi = k$), any zero-error scheme satisfies $L \geq \log_2 k$.*

(L: LWDC3)

D. Witness Cost for Class Identity

Recall from the model setup that the witness cost $W(P)$ is the minimum number of primitive queries required to compute property P. For class identity, we ask: what is the minimum number of queries to determine if two values have the same class?

Theorem III.15 (Nominal-Tag Observers Achieve Minimum Witness Cost)**.** *Nominal-tag observers achieve the minimum witness cost for class identity:*

$$W_{eq} = O(1).$$

Specifically, the witness is a single tag comparison. Attribute-only observers require $W_{eq} = \Omega(d)$ where d is the distinguishing dimension (and $d \leq n$, with worst-case $d = n$).

(L: ACS7, ACS1)

Proof.

1) Nominal-tag access is a single primitive query.
2) Attribute-only observers must query at least d attributes in the worst case (a generic strategy queries all n).
3) No shorter witness exists for attribute-only observers by the information barrier and the distinguishing-dimension lower bound.

■

The supplementary artifact provides a machine-checked proof that nominal-tag access minimizes witness cost for class identity.

IV. Query Complexity and the Structure of Distinguishing Sets

A. Model Contract (Fixed-Axis Domains)

Model contract (fixed-axis domain). A domain is specified by a fixed observation family Φ derived from a fixed axis map $\alpha : \mathcal{V} \to \mathcal{A}$ (e.g., $\alpha(v) = (B(v), S(v))$). An observer is permitted to interact with v only through primitive queries in Φ, and each primitive query factors through α: for every $q \in \Phi$, there exists $\tilde{q}$ such that $q(v) = \tilde{q}(\alpha(v))$. A property is in-scope semantic iff it is computable by an admissible strategy that uses only responses to queries in Φ.

We adopt Φ as the complete observation universe for this paper: to claim applicability to a concrete runtime one must either (i) exhibit mappings from each runtime observable into Φ, or (ii) enforce the admissibility constraints (no external registries, no reflection, no preprocessing or amortization). Under either condition the theorems apply without qualification.

The fixed-axis contract means the observer works on an observational quotient: all in-scope semantic properties factor through α, and any mechanism that distinguishes values inside an α-fiber must introduce additional information not present in the declared observation family.

B. Query Families and Distinguishing Sets

The classification problem is: given a set of queries, which subsets suffice to distinguish all entities?

Definition IV.1 (Query family)**.** Let $\mathcal{Q}$ be the set of all primitive queries available to an observer. For a classification system with attribute set $\mathcal{I}$, we have $\mathcal{Q} = \{q_I : I \in \mathcal{I}\}$ where $q_I(v) = 1$ iff v satisfies attribute I.

(L: ACS1)

In this section, "queries" are the primitive attribute predicates $q \in \Phi$ (equivalently, each q factors through the axis map: $q = \tilde{q} \circ \alpha$).

Convention: $\Phi := \mathcal{Q}$. All universal quantification over "queries" ranges over $q \in \Phi$ only.

Definition IV.2 (Distinguishing set)**.** A subset $S \subseteq \mathcal{Q}$ is *distinguishing* if, for all values v, w with $\mathrm{class}(v) \neq \mathrm{class}(w)$, there exists $q \in S$ such that $q(v) \neq q(w)$.

(L: L4, L5)

Definition IV.3 (Minimal distinguishing set)**.** A distinguishing set S is *minimal* if no proper subset of S is distinguishing.

(L: L1, L4)

C. Matroid Structure of Query Families

Scope and assumptions. The matroid theorem below is unconditional within the fixed-axis observational theory defined above. In this section, "query" always means a primitive predicate $q \in \Phi$. It depends only on:

- $E = \Phi$ is the ground set of primitive queries (attribute predicates).
- "Distinguishing": for all values v, w with $\mathrm{class}(v) \neq \mathrm{class}(w)$, there exists $q \in S$ such that $q(v) \neq q(w)$.
- "Minimal" means inclusion-minimal: no proper subset suffices.

No further assumptions are required within this theory beyond the fixed observation family Φ. The mechanized development verifies the closure laws, the exchange and equicardinality lemmas, and the closure-to-matroid bridge used below; the representative handle tags are L1, L4, and L5.

Definition IV.4 (Bases family)**.** Let $E = \Phi$ $(= \mathcal{Q})$ be the ground set of primitive queries. Let $\mathcal{B} \subseteq 2^E$ be the family of minimal distinguishing sets.

(L: L1, L5)

Lemma IV.5 (Basis exchange)**.** *For any $B_1, B_2 \in \mathcal{B}$ and any $q \in B_1 \setminus B_2$, there exists $q' \in B_2 \setminus B_1$ such that $(B_1 \setminus \{q\}) \cup \{q'\} \in \mathcal{B}$.*

(L: L4, L5)

Proof. Define the closure operator $\mathrm{cl}(X) = \{q : X\text{-equivalence implies } q\text{-equivalence}\}$. We verify the matroid axioms:

1) **Closure axioms**: cl is extensive, monotone, and idempotent. These follow directly from the definition of logical implication.
2) **Exchange property**: If $q \in \mathrm{cl}(X \cup \{q'\}) \setminus \mathrm{cl}(X)$, then $q' \in \mathrm{cl}(X \cup \{q\})$.

For exchange, take $q \in \mathrm{cl}(X \cup \{q'\}) \setminus \mathrm{cl}(X)$. Since $q \notin \mathrm{cl}(X)$, there exist v, w that are X-equivalent but disagree on q. Because $q \in \mathrm{cl}(X \cup \{q'\})$, any pair that is $(X \cup \{q'\})$-equivalent must agree on q; therefore this witness pair cannot be $(X \cup \{q'\})$-equivalent, so it must disagree on q'. Now fix any pair v', w' that are $(X \cup \{q\})$-equivalent. They are in particular X-equivalent and agree on q. If they disagreed on q', then by the previous implication we could derive disagreement on q, contradiction. Hence v', w' agree on q', proving $q' \in \mathrm{cl}(X \cup \{q\})$.

Minimal distinguishing sets are exactly the bases of the matroid defined by this closure operator. The machine-checked development verifies the closure laws, the exchange and equicardinality lemmas, and the bridge from closure to matroid structure used to make the distinguishing dimension well-defined. ■

Theorem IV.6 (Matroid bases)**.** *$\mathcal{B}$ is the set of bases of a matroid on ground set E.*

(L: L1, L5)

Proof. By the basis-exchange lemma and the standard characterization of matroid bases [12]. ■

Definition IV.7 (Distinguishing dimension)**.** The *distinguishing dimension* of a classification system is the common cardinality of all minimal distinguishing sets.

(L: L1)

Remark IV.8 (Ambient attribute count vs. distinguishing dimension). Let $n := |\mathcal{I}|$ be the ambient number of available attributes. Clearly $d \leq n$, and there exist worst-case families with $d = n$.

(L: ACS1, L1)

Corollary IV.9 (Well-defined distinguishing dimension)**.** *All minimal distinguishing sets have equal cardinality. Thus the distinguishing dimension is well-defined.*

(L: L1)

D. Implications for Witness Cost

Theorem IV.10 (Attribute-only lower bound)**.** *For attribute-only observers, class identity checking requires*

$$W(\textit{class-identity}) = \Omega(d)$$

in the worst case, where d is the distinguishing dimension.

(L: ACS1)

Proof. Assume a zero-error attribute-only procedure halts after fewer than d queries on every execution path. Fix any execution path and let $Q \subseteq \mathcal{I}$ be the set of queried attributes on that path, so $|Q| < d$. Since d is the cardinality of every minimal distinguishing set, no set of size $< d$ is distinguishing; hence there exist values v, w from different classes with identical answers on all attributes in Q.

An adversary can answer the procedure's queries consistently with both v and w along this path. Therefore the resulting transcript (and output) is identical on v and w, contradicting zero-error class identification. So some execution path must use at least d queries, giving worst-case cost $\Omega(d)$. ■

Corollary IV.11 (Lower bound on attribute-only witness cost)**.** *For any attribute-only observer, W(class-identity) $\geq d$ where d is the distinguishing dimension.*

(L: ACS1)

The key insight: the distinguishing dimension is invariant across all minimal query strategies. The difference between nominal-tag and attribute-only observers lies in *witness cost*: a nominal tag achieves $W = O(1)$ by storing the identity directly, bypassing query enumeration.

E. Witness Cost Comparison

Observer Class	**Witness Procedure**	**Witness Cost** W
Nominal-tag	Single tag read	$O(1)$
Attribute-only	Query a distinguishing set	$\Omega(d)$

TABLE I
WITNESS COST FOR CLASS IDENTITY BY OBSERVER CLASS.

V. (L, W, D) FRONTIER AND EXACT FINITE-BLOCK SCALING

A. Three-Dimensional Tradeoff: Tag Length, Witness Cost, Distortion

Recall from the preceding sections that observer strategies are characterized by three dimensions:

- **Tag length** L: bits required to encode class information,
- **Witness cost** W: minimum number of primitive queries for class identification,
- **Distortion** D: probability of misclassification, $D = \Pr[\hat{C} \neq C]$.

We compare two observer classes: attribute-only observers, which query only attribute membership, and nominal-tag observers, which may read a class identifier in addition to attribute queries.

Theorem V.1 (Pareto Optimality of Nominal-Tag Observers)**.** *Let A_π be the collision multiplicity (Definition III.11). Then:*

1) *Any $D = 0$ scheme must satisfy $L \geq \log_2 A_\pi$ (Theorem III.13).*
2) *In maximal-barrier domains ($A_\pi = k$), nominal-tag observers achieve the unique Pareto-optimal $D = 0$ point:*
   - ***Tag length**: $L = \lceil \log_2 k \rceil$ bits,*
   - ***Witness cost**: $W = O(1)$ queries,*
   - ***Distortion**: $D = 0$.*
3) *In general (non-maximal) domains, nominal tagging remains Pareto-optimal at $D = 0$ but need not be unique: partial tags can coexist on the frontier.*

In information-barrier domains, attribute-only observers (the $L = 0$ face) satisfy:

- ***Tag length**: $L = 0$ bits,*
- ***Witness cost**: $W = \Omega(d)$ queries,*
- ***Distortion**: $D > 0$ when collisions persist.*

(L: LWDC1, LWDC3, LWDC2)

Proof. Converse item (1) is Theorem III.13. For item (2), maximal barrier means all classes are observationally colliding, so any $D = 0$ scheme must carry full class identity in tag bits (Corollary III.14), while nominal tags realize this lower bound with one tag read. Pareto uniqueness follows because any competing $D = 0$ point cannot reduce L below $\log_2 k$ nor reduce W below constant-time tag access under the admissibility rules of Section II.

The converse chain has two steps: first, a constant transcript on a collision block forces any zero-error decoder to separate classes using tag information; second, counting those tag outcomes yields $2^L \geq A_\pi$. In the maximal-barrier regime this becomes $L \geq \log_2 k$, which nominal tagging achieves with one tag read. ∎

Remark V.2 (General $D = 0$ Frontier). When $1 < A_\pi < k$, the $D = 0$ frontier can include mixed designs: a partial tag identifies collision blocks and queries resolve within blocks. This

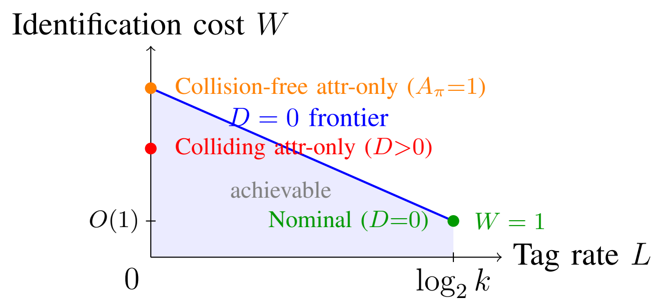

Fig. 1. Schematic (L, W, D) tradeoff across barrier regimes. Here W counts identification operations, including a constant-time tag read, so the nominal zero-error point sits at $W = O(1)$ rather than $W = 0$. The left orange point shows the collision-free attribute-only regime ($A_\pi = 1$), while the red point shows a colliding attribute-only regime with $D > 0$. Nominal tagging is the unique zero-error Pareto point in maximal-barrier domains.

does not contradict nominal optimality; it only removes global uniqueness outside maximal-barrier domains.

1) Maximal barrier ($A_\pi = k$): unique $D = 0$ nominal point.
2) Intermediate barrier ($1 < A_\pi < k$): multiple $D = 0$ Pareto points may exist.
3) No barrier ($A_\pi = 1$): $L = 0$ zero-error identification is feasible.

(L: LWDC1, LWDC2, LWDC3)

B. Pareto Frontier

The three-dimensional frontier shows:

- in maximal-barrier domains, the unique $D = 0$ Pareto point is nominal tagging at $L = \lceil \log_2 k \rceil$,
- in general domains, attribute-only observers trade tag length for distortion on the $L = 0$ face when collisions are present.

Figure 1 visualizes the (L, W, D) tradeoff space. The key observation is the ambiguity converse: the minimum zero-error tag rate is $\log_2 A_\pi$, with maximal-barrier specialization $\log_2 k$.

The supplementary artifact machine-checks the full ambiguity-based converse chain and maximal-barrier lower bound that anchor this tradeoff analysis.

Remark V.3 (Software-runtime instantiation (example)). In one software-runtime instantiation: explicit runtime identifiers correspond to nominal-tag observers (e.g., CPython's `isinstance`, Java's `.getClass()`); attribute-probe schemes correspond to attribute-only observers (e.g., repeated `hasattr`-style checks); and interface-conformance checks are an intermediate case with $D = 0$ but $W = O(s)$.

(L: ACS7, ACS1, FORC2)

Remark V.4 (Conformance-check cost parameter). When conformance checks traverse s members or fields (rather than ranging over the full attribute universe), the natural bound is $W = O(s)$ with $s \leq n$.

(L: ACS1)

C. Confusability Graph Thresholds and Block Scaling

Let $\pi_{\mathcal{C}} : \mathcal{C} \to \{0,1\}^n$ be the class-profile map. The induced confusability graph connects two distinct classes when they share the same observable profile.

Theorem V.5 (Exact one-shot feasibility threshold)**.** *For an alphabet of n nominal tags, zero-error class tagging is feasible if and only if*

$$A_\pi \leq n.$$

(L: GPH13)

Proof. Classes in the same profile fiber are pairwise confusable, so each such fiber is a clique in the confusability graph. Zero-error tagging is equivalent to a proper graph coloring; therefore feasibility is equivalent to having at least as many colors as the largest clique size, which is exactly A_π. ■

Theorem V.6 (Exact block feasibility law)**.** *For block length $t \geq 1$ with coordinatewise observation on $\mathcal{C}^t$, zero-error tagging with alphabet size n_t is feasible if and only if*

$$A_\pi^t \leq n_t.$$

Equivalently, the minimal feasible block alphabet is exactly A_π^t.

(L: GPH16, GPH17, GPH20)

Corollary V.7 (Linear log-bit scaling). *Let $L_t^\star := \log_2(A_\pi^t)$ be the minimum block tag budget in bits. Then*

$$L_t^\star = t\,\log_2 A_\pi, \qquad \frac{L_t^\star}{t} = \log_2 A_\pi.$$

So per-entity tag rate is exactly stable across block length.

(L: GPH18, GPH19, GPH24)

Remark V.8 (Scope). These are worst-case finite-block confusability laws. They strengthen the zero-error converse with exact block thresholds, while remaining distinct from full distribution-optimized Körner graph-entropy theory.

(L: GPH13, GPH16, GPH18, GPH19, GPH24)

VI. Robustness and Limits of Collision-Free Guarantees

A. Open-World Robustness and Decidability Boundary

The main converse can be challenged by saying that the current realized world happens to be collision-free, so the lower bound is irrelevant in practice. The next two theorems show why that defense is not stable.

Definition VI.1 (Finite realized world). Let $\mathcal{C}$ be the class universe. A *finite realized world* is a finite subset $W \subseteq \mathcal{C}$ of classes currently present in a deployed system snapshot. We call W *barrier-free* iff

$$\forall c_1, c_2 \in W,\ \pi_{\mathcal{C}}(c_1) = \pi_{\mathcal{C}}(c_2) \Rightarrow c_1 = c_2.$$

(L: ROB1)

Definition VI.2 (Open-world extension). An *open-world extension* is a super-world $W' \supseteq W$ obtained by adding classes while keeping the observation family Φ fixed.

(L: ROB1, ROB2)

Theorem VI.3 (Barrier-freedom is not extension-stable)**.** *In the open-world class model formalized in Lean, it is not true that barrier-freedom is preserved under all extensions:*

$$\neg\Big(\forall W \subseteq W',\ \mathrm{BarrierFree}(W) \Rightarrow \mathrm{BarrierFree}(W')\Big).$$

(L: ROB1, ROB2)

Proof. Take the empty world $W_0 = \varnothing$, which is barrier-free. In the Lean model, two concrete classes are shape-equivalent (same observable profile) but nominally distinct, so adding them yields an extension $W_1 \supseteq W_0$ that is not barrier-free. Therefore barrier-freedom is not extension-stable. ■

Definition VI.4 (Function-level barrier predicate)**.** For a partial observer generator $f : \mathbb{N} \rightharpoonup \mathbb{N}$, define

$$\mathrm{HasBarrier}(f) \ :\Leftrightarrow\ \exists x \neq y, \exists z,\ z \in f(x) \wedge z \in f(y).$$

Define $\mathrm{BarrierFree}(f) :\Leftrightarrow \neg\mathrm{HasBarrier}(f)$.

(L: UND1, UND2)

Theorem VI.5 (Rice-style non-computability of barrier certification)**.** *There is no computable predicate on program codes deciding whether the generated observer has a barrier:*

$$\neg\mathrm{ComputablePred}(c \mapsto \mathrm{HasBarrier}(\mathrm{eval}(c)))\,.$$

Equivalently, barrier-freedom certification is also non-computable.

(L: UND1, UND2)

Proof. Assume there is a computable certifier for $P(c) := \mathrm{HasBarrier}(\mathrm{eval}(c))$. The property is non-trivial: some generators produce collisions and some do not. Rice's theorem therefore implies that if membership in the corresponding semantic class were computable from code, every partial recursive function would have to satisfy it, contradiction. The barrier-free statement is equivalent by negation. ■

TABLE II
IDENTIFICATION STRATEGIES FOR 1000 CLASSES WITH 50 ATTRIBUTES.

Strategy	**Tag** L	**Witness** W
Explicit-tag identification	$\lceil \log_2 1000 \rceil = 10$ bits	$O(1)$
Attribute-only (exhaustive)	0	≤ 50 queries
Attribute-only (adaptive)	0	$\geq d$ queries

Corollary VI.6 (Certification consequence for persistent $D = 0$ claims)**.** *For unrestricted open-world generators, one cannot computably certify that attribute-only identification remains barrier-free under all future extensions. Consequently, persistent $D = 0$ guarantees cannot rely on "currently collision-free" attribute structure alone.*

(L: ROB1, ROB2, UND1, UND2)

VII. Practical Consequences and Cross-Domain Illustrations

A. Concrete Example

Consider a classification system with $k = 1000$ classes, each characterized by a subset of $n = 50$ binary attributes. Table II compares the strategies.

Here d is the distinguishing dimension, the size of any minimal distinguishing query set. For many practical systems, d is well below n but nonzero. The gap between 10 bits of storage versus 5–50 queries per identification is the cost of forgoing explicit tags.

B. The Error Localization Theorem

a) Scope of this subsection.: This subsection studies *maintenance or localization* complexity, measured by locations to inspect ($E(\cdot)$), not per-instance identification complexity (W) from earlier sections. The metrics are related but distinct: W quantifies online class-identification effort, while E quantifies where constraint logic is distributed in implementations.

Definition VII.1 (Error location count)**.** Let $E(\mathcal{O})$ be the number of locations that must be inspected to find all potential violations of a constraint under observation family $\mathcal{O}$.

(L: ACS3, ACS7)

Theorem VII.2 (Nominal-tag localization)**.** $E(\textit{nominal-tag}) = O(1)$.

(L: ACS7, DOM1, FORC2)

Proof. Under nominal-tag observation, the constraint "v must be of class A" is satisfied iff $\tau(v) \in \text{subtypes}(A)$. This is determined at a single location: the definition of $\tau(v)$'s class. ■

Theorem VII.3 (Declared-attribute localization)**.** *$E(\textit{attribute-only, declared}) = O(k)$ where k is the number of entity classes.*

(L: ACS3)

Proof. With declared attribute sets, the constraint "v must satisfy attribute I" requires verifying that each class satisfies all attributes in I. For k classes, this yields $O(k)$ locations. ■

Theorem VII.4 (Attribute-only localization)**.** *$E(\textit{attribute-only}) = \Omega(m)$ where m is the number of query sites.*

(L: ACS3, ACS9, COH1)

Proof. Under attribute-only observation, each query site independently checks "does v have attribute a?" with no centralized declaration. For m query sites, each must be inspected. The lower bound is therefore $\Omega(m)$. ■

Corollary VII.5 (Strict dominance)**.** *Nominal-tag observation strictly dominates attribute-only: $E(\textit{nominal-tag}) = O(1) < \Omega(m) = E(\textit{attribute-only})$ for all $m > 1$.*

(L: ACS4, DOM1, FORC2)

C. The Information Scattering Theorem

Definition VII.6 (Constraint encoding locations)**.** Let $I(\mathcal{O}, c)$ be the set of locations where constraint c is encoded under observation family $\mathcal{O}$.

(L: ACS3, ACS6)

Theorem VII.7 (Attribute-only scattering)**.** *For attribute-only observation, $|I(\textit{attribute-only}, c)| = O(m)$ where m is the number of query sites using constraint c.*

(L: ACS3)

Proof. Each attribute query independently encodes the constraint. No shared reference exists. Constraint encodings therefore scale with query sites. ∎

Theorem VII.8 (Nominal-tag centralization)**.** *For nominal-tag observation, $|I(\textit{nominal-tag}, c)| = O(1)$.*

(L: ACS6, UNIQ1, DOM1)

Proof. The constraint "must be of class A" is encoded once in the definition of A. All tag checks reference this single definition. ∎

Corollary VII.9 (Maintenance entropy)**.** *Attribute-only observation maximizes maintenance entropy; nominal-tag observation minimizes it.*

(L: ACS2, COH1, FORC2)

D. Compact Cross-Domain Illustrations

Biology (phenotype vs genotype). Distinct species can collide under phenotype, so phenotype-only observation cannot guarantee zero-error identity. DNA barcoding supplies additional naming information that resolves collision blocks in constant-time lookup style [13].

Databases (columns vs keys). Rows may collide on non-key columns, so attribute-only identification is insufficient for guaranteed identity. Primary or surrogate keys realize the nominal-tag role and provide constant-cost identity checks [14].

Software runtimes (attribute probes vs runtime identifiers). Attribute-probe checks recover identity from observed capabilities and pay query cost that scales with inspected structure. Runtime class or type identifiers behave as explicit tags and collapse identity checks to near-constant cost in the model [15]–[19].

Model registries and artifact stores. Structural signatures can collide across distinct artifacts. Registry identifiers and hashes play the same nominal role as tags, while profile inspection alone inherits the ambiguity lower bounds.

E. Takeaway

These illustrations are not historical causality claims. They show a common resource law: when observable profiles collide, zero-error identification requires additional naming information; otherwise the system pays higher query cost, higher maintenance complexity, and/or nonzero error.

The examples are consequences of the main theorem family rather than separate stories. Across domains, the recurring design constraint is the same: once observable profiles collide, systems must spend either explicit naming bits or additional structure-specific work, and persistent guarantees are strongest when the naming information is carried directly.

VIII. Conclusion

This paper presents a resource-based analysis of classification under observational constraints. The central law is the zero-error ambiguity converse: any $D = 0$ scheme must satisfy $L \geq \log_2 A_\pi$, where A_π is the largest collision block induced by observable profiles. The finite-block confusability results sharpen that law to exact thresholds and exact linear log-bit scaling across block length. In maximal-barrier domains, nominal tagging achieves the unique Pareto-optimal zero-error point in the (L, W, D) tradeoff.

The second main arc characterizes the query resource. Minimal distinguishing query sets form the bases of a matroid, making the distinguishing dimension d well-defined as an invariant of the observation structure. Attribute-only zero-error identification therefore pays at least d queries in the worst case. Open-world and Rice-style boundary results explain why present-day collision-freedom is not enough for persistent guarantees, and the implementation-level results show how the same structure reappears as maintenance localization and information scattering.

A. A Recurring Design Constraint

Across domains, the same structure recurs:

- **Biology**: phenotypic observation cannot distinguish cryptic species. DNA barcoding supplies additional naming information.
- **Databases**: column-value queries cannot distinguish rows with identical observable attributes. Primary keys provide explicit identity.
- **Software runtimes**: attribute observation cannot distinguish entities that collide on observed capabilities. Runtime identifiers provide constant-cost identity.
- **Model registries**: structural signatures can collide, and registry identifiers play the same nominal role as tags.

The information barrier is not a quirk of any particular domain; it is a mathematical consequence of the quotient structure induced by limited observations. What varies by domain is not the existence of the constraint, but which resource the system chooses to spend to escape it.

B. Implications

- **The necessity of nominal information is a theorem, not a preference.** Any zero-error scheme must satisfy the ambiguity converse $L \geq \log_2 A_\pi$ (Theorem III.13). In maximal-barrier domains ($A_\pi = k$), this becomes $L \geq \log_2 k$, and nominal tagging gives the unique $D = 0$ Pareto point with $W = O(1)$ (Theorem V.1).
- **The barrier is informational, not computational.** Even with unbounded resources, attribute-only observers cannot overcome it.
- **Finite-block confusability laws are exact.** The graph-confusability results give one-shot and block thresholds (A_π and A_π^t) and exact linear log-bit scaling across block length.
- **Open-world guarantees require explicit tags.** Barrier-freedom is not extension-stable under system growth, and deciding barrier existence or freedom for arbitrary generators is not computable. So a persistent $D = 0$ guarantee cannot rely on currently collision-free attribute structure alone.

- **Fixed-axis systems are necessarily incomplete outside their axis.** By Corollary II.16, any fixed-axis classifier is complete only for axis-measurable properties and cannot represent properties that vary within an axis fiber unless a new axis or explicit tag is introduced.

(L: ROB1, ROB2, UND1, UND2)

C. Future Work

Natural extensions include other classification domains, witness complexity for semantic properties beyond identity, and hybrid observers that combine tags and attributes under non-uniform query distributions.

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Generative AI tools (including Codex, Claude Code, Augment, Kilo, and OpenCode) were used throughout this manuscript, across all sections (Abstract, Introduction, theoretical development, proof sketches, applications, conclusion, and appendix) and across all stages from initial drafting to final revision. The tools were used for boilerplate generation, prose and notation refinement, LaTeX and structure cleanup, translation of informal proof ideas into candidate formal artifacts (Lean and LaTeX), and repeated adversarial reviewer-style critique passes to identify blind spots and clarity gaps.

The author retained full intellectual and editorial control, including problem selection, theorem statements, assumptions, novelty framing, acceptance criteria, and final inclusion or exclusion decisions. No technical claim was accepted solely from AI output. Formal claims reported as machine-verified were admitted only after Lean verification (no `sorry` in cited modules) and direct author review; Lean was used as an integrity gate for responsible AI-assisted research. The author is solely responsible for all statements, citations, and conclusions.

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Lean Handle Index.

ID	Handle	ID	Handle
ACS1	adversary_forces_n_minus_1_queries `AbstractClassSystem/Extended.lean`	ACS2	complexity_gap_unbounded `AbstractClassSystem/Extended.lean`
ACS3	duck_localization_linear `AbstractClassSystem/Extended.lean`	ACS4	encoding_location_gap `AbstractClassSystem/Extended.lean`
ACS5	model_completeness `AbstractClassSystem/Extended.lean`	ACS6	nominal_centralized `AbstractClassSystem/Extended.lean`

ID	Handle	ID	Handle
ACS7	nominal_localization_constant_semantic `AbstractClassSystem/Extended.lean`	ACS8	shape_cannot_distinguish `AbstractClassSystem/Core.lean`
ACS9	shape_provenance_impossible `AbstractClassSystem/Core.lean`	COH1	preference_incoherent `AbstractClassSystem/Bridge.lean`
COMPL1	axes_complete `AbstractClassSystem/Extended.lean`	DOM1	strict_dominance `AbstractClassSystem/Core.lean`
FORC2	typing_choice_unavoidable `AbstractClassSystem/Bridge.lean`	GPH13	Ssot.GraphEntropy.tagFeasible_iff_maxFiberCard_le
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UND2	AbstractClassSystem.OpenWorld.barrierFree_not_computable `AbstractClassSystem/Undecidability.lean`	UNIQ1	unique_tree_encoding `AbstractClassSystem/Extended.lean`